

Robot Operating System (ROS) – Setup

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PC Setup

- **Sourec: <https://wiki.ros.org/noetic/Installation/Ubuntu>**
- Previous class we installed Ubuntu 20.04
- After installation do some setups:
 - `$ sudo apt-get update && sudo apt-get upgrade`
 - **Setup your sources.list**
 - `$ sudo sh -c 'echo "deb http://packages.ros.org/ros/ubuntu $(lsb_release -sc) main" > /etc/apt/sources.list.d/ros-latest.list'`
 - **Set up your keys**
 - `$ sudo apt install curl` # if you haven't already installed curl
 - `$ curl -s https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc | sudo apt-key add -`
 - **Make sure your Debian package index is up-to-date**
 - `$ sudo apt update`

Installing ROS on PC

- **Desktop-Full Install: (Recommended)**
- `$ sudo apt install ros-noetic-desktop-full`
- rqt installation –
- `$ sudo apt-get install ros-noetic-rqt*`
- Individual package installation:
 - `$ sudo apt install ros-noetic-PACKAGE`
 - e.g. `$ sudo apt install ros-noetic-slam-gmapping`
- **To find available packages, use:**
 - `$ apt search ros-noetic`

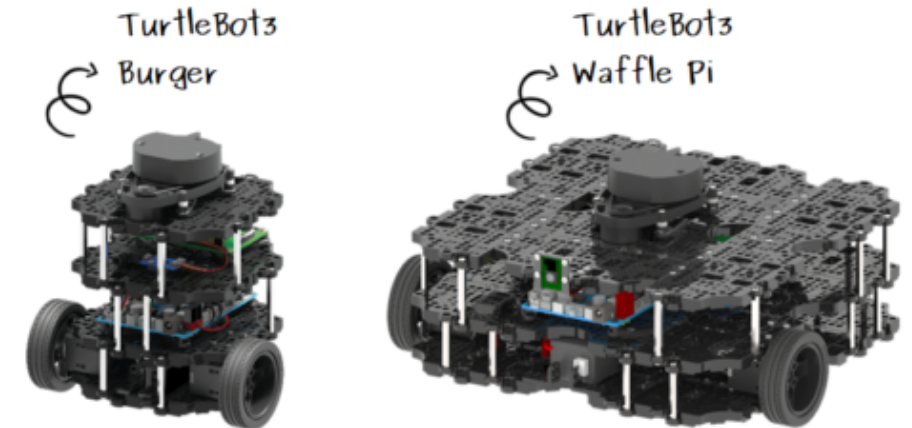
Environment setup

- ROS environment variables are automatically added to your bash session:
 - `$ echo "source /opt/ros/noetic/setup.bash" >> ~/.bashrc`
 - `$ source ~/.bashrc`
- change the environment of your current shell:
 - `$ source /opt/ros/noetic/setup.bash`
- Dependencies for building packages:
 - `$ sudo apt install python-rosdep python-rosinstall python-rosinstall-generator python-wstool build-essential`
- Initialize rosdep:
 - `$ sudo apt install python-rosdep`
 - `$ sudo rosdep init`
 - `$ rosdep update`

Hardware setup

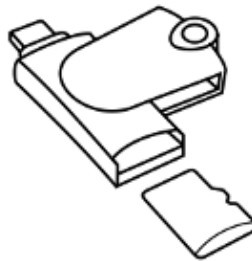
Hardware Assembly

- TURTLEBOT3_Burger_QS_Ver1815_low(web).pdf >>> <https://www.robotis.com/service/download.php?no=748>
- TURTLEBOT3_Waffle-Pi_QS_Ver1916(low).pdf >>> <https://www.robotis.com/service/download.php?no=750>



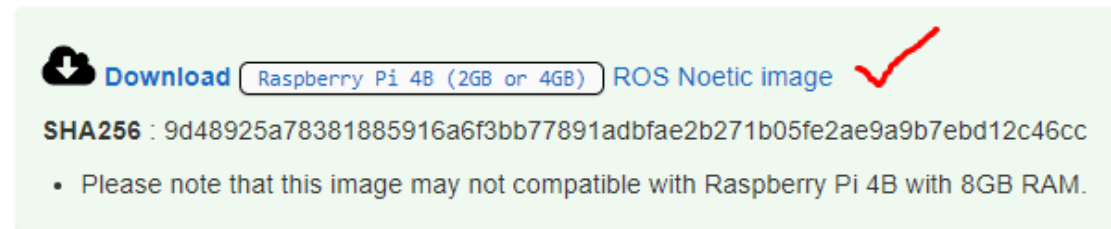
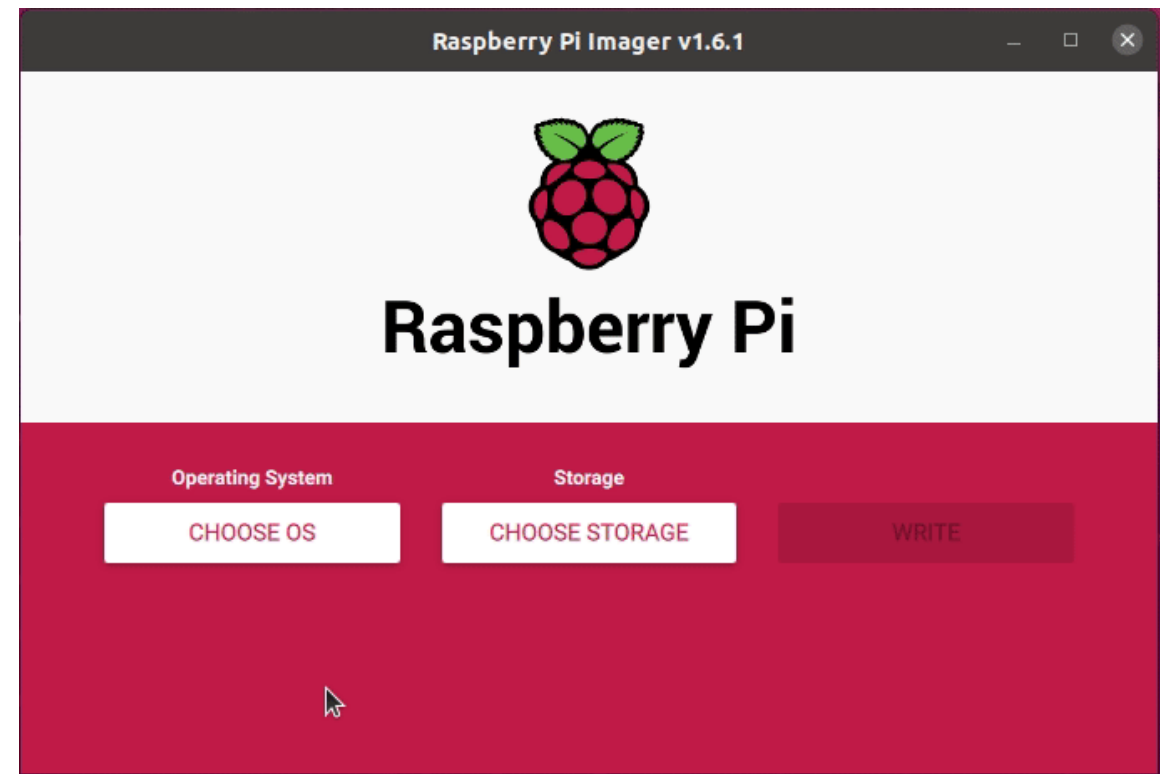
Turtlebot3 Setup

- Prepare microSD Card and Reader
- Download TurtleBot3 SBC Image (choose Noetic)
 - https://emanual.robotis.com/docs/en/platform/turtlebot3/sbc_setup/#sbc-setup
- Then, follow the instructions on the SBC (Raspberry Pi) of TurtleBot3...



Raspberry Pi Imager

<https://www.raspberrypi.com/software/>



<https://www.robotis.com/service/download.php?no=2066>

Turtlebot3 Setup

- Boot Up the Raspberry Pi
 - a) Connect the HDMI cable of the monitor to the HDMI port of Raspberry Pi.
 - b) Connect input devices to the USB port of Raspberry Pi.
 - c) Insert the microSD card.
 - d) Connect the power (either with USB or OpenCR) to turn on the Raspberry Pi.
- Configure the Raspberry Pi
 - a) Log in with default username (ubuntu) and password ('turtlebot' or 'ubuntu'). After logged in, system will ask you to change the password.
 - b) Open automatic update setting file.
\$ sudo nano /etc/apt/apt.conf.d/20auto-upgrades
- Change the update settings as below
APT::Periodic::Update-Package-Lists "0";
APT::Periodic::Unattended-Upgrade "0";
 - a. Save the file with '**Ctrl+S**' and exit with '**Ctrl+X**'

Turtlebot3 Setup

- Enter below command to configure the WiFi network setting
 - Replace the WIFI_SSID and WIFI_PASSWORD with your wifi SSID and password
 - Save the file with 'Ctrl+S' and exit with 'Ctrl+X'
- Reboot the Raspberry Pi
 - **\$ sudo reboot**
- Set the systemd to prevent boot-up delay even if there is no network at startup. Run the command below to set mask the systemd process using the following command
\$ systemctl mask systemd-networkd-wait-online.service
- Disable Suspend and Hibernation
\$ sudo systemctl mask sleep.target suspend.target hibernate.target hybrid-sleep.target
- After rebooting the Raspberry Pi, if you wish to work from the Remote PC using SSH, use below command from the remote PC terminal. The default password is **ubuntu** or **turtlebot**.
 - **\$ ssh ubuntu@{IP Address of Raspberry PI}**

```
network:
  version: 2
  renderer: networkd
  ethernets:
    eth0:
      dhcp4: yes
      dhcp6: yes
      optional: true
  wifis:
    wlan0:
      dhcp4: yes
      dhcp6: yes
      access-points:
        WIFI_SSID:
          password: WIFI_PASSWORD
```

Turtlebot3 Setup

- Install ROS Noetic Ninjemys Enter below commands to the terminal one at a time

```
$ sudo apt-get update
```

```
$ sudo apt-get upgrade
```

```
$ wget https://raw.githubusercontent.com/ROBOTIS-GIT/  
robotis_tools/master/install_ros_noetic_rpi.sh
```

```
$ chmod 755 ./install_ros_noetic_rpi.sh
```

```
$ bash ./install_ros_noetic_rpi.sh
```

- If the above installation fails, please refer to the [official ROS Noetic installation guide](#).

Turtlebot3 Setup

- Install and Build ROS Packages

```
$ sudo apt install ros-noetic-rosserial-python ros-noetic-tf
```

```
$ sudo apt install ros-noetic-hls-lfcd-lds-driver
```

```
$ sudo apt install ros-noetic-turtlebot3-msgs
```

```
$ sudo apt install ros-noetic-dynamixel-sdk
```

```
$ cd ~/catkin_ws/src
```

```
$ git clone -b noetic-devel https://github.com/ROBOTIS-GIT/turtlebot3.git
```

```
$ cd ~/catkin_ws/src/turtlebot3
```

```
$ rm -r turtlebot3_description/ turtlebot3_teleop/ turtlebot3_navigation/  
turtlebot3_slam/ turtlebot3_example/
```

```
$ cd ~/catkin_ws/
```

```
$ catkin_make -j1
```

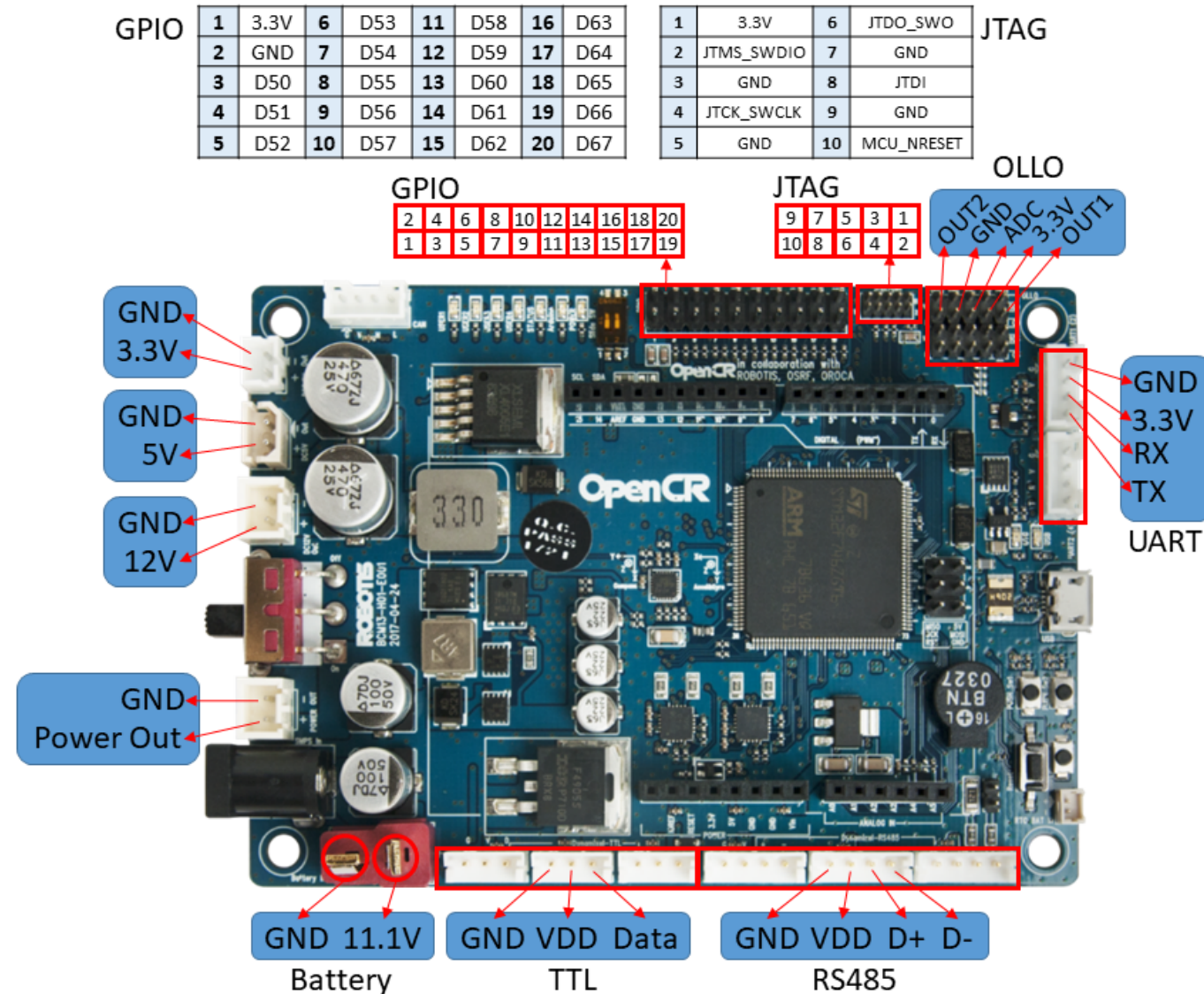
```
$ source ~/.bashrc
```

Turtlebot3 Setup

- USB Port Setting
\$ **roslaunch turtlebot3_bringup create_udev_rules**
- ROS Network Configuration Confirm the WiFi IP address and edit the **.bashrc** file
\$ **nano ~/.bashrc**
- Modify the IP addresses of **ROS_MASTER_URI** and the **ROS_HOSTNAME**
export ROS_MASTER_URI=http://{IP_ADDRESS_OF_REMOTE_PC}:11311
export ROS_HOSTNAME={IP_ADDRESS_OF_RASPBERRY_PI_3}
 - Save the file and exit the nano editor

OpenCR

- OpenCR1.0 is developed for ROS embedded systems to provide completely open-source hardware and software.



OpenCR

- Connect the OpenCR to the Raspberry Pi using the micro USB cable.
- Install required packages on the Raspberry Pi to upload the OpenCR firmware.
 - `$ sudo dpkg --add-architecture armhf`
 - `$ sudo apt-get update`
 - `$ sudo apt-get install libc6:armhf`
- Depending on the platform, use either 'burger' or 'waffle_pi' for the OPENCR_MODEL name.
 - `$ export OPENCR_PORT=/dev/ttyACM0`
 - `$ export OPENCR_MODEL=burger_noetic`
 - `$ rm -rf ./opencr_update.tar.bz2`
- Download the firmware and loader, then extract the file.
 - `$ wget https://github.com/ROBOTIS-GIT/OpenCR-Binaries/raw/master/turtlebot3/ROS1/latest/opencr_update.tar.bz2`
 - `$ tar -xvf opencr_update.tar.bz2`
- Upload firmware to the OpenCR.
 - `$ cd ./opencr_update`
 - `$ ./update.sh $OPENCRCR_PORT $OPENCRCR_MODEL.opencr`

OpenCR

- A successful firmware upload for TurtleBot3 Burger will look like below.
- Click here to expand more details!
- https://emanual.robotis.com/docs/en/platform/turtlebot3/opencr_setup/#opencr-setup

```
turtlebot@turtlebot: ~  
turtlebot@turtlebot: ~ 98x55  
turtlebot@turtlebot:~$ export OPENCN_PORT=/dev/ttyACM0  
turtlebot@turtlebot:~$ export OPENCN_MODEL=burger  
turtlebot@turtlebot:~$ rm -rf ./opencr_update.tar.bz2  
turtlebot@turtlebot:~$ wget https://github.com/ROBOTIS-GIT/OpenCR/raw/develop/arduino/opencr_release/shell_update/opencr_update.tar.bz2 && tar -xvf opencr_update.tar.bz2 && cd ./opencr_update && ./update.sh $OPENCN_PORT $OPENCN_MODEL.opencr && cd ..  
--2018-03-05 11:42:25-- https://github.com/ROBOTIS-GIT/OpenCR/raw/develop/arduino/opencr_release/shell_update/opencr_update.tar.bz2  
Resolving github.com (github.com)... 192.30.255.112, 192.30.255.113  
Connecting to github.com (github.com)|192.30.255.112|:443... connected.  
HTTP request sent, awaiting response... 302 Found  
Location: https://raw.githubusercontent.com/ROBOTIS-GIT/OpenCR/develop/arduino/opencr_release/shell_update/opencr_update.tar.bz2 [following]  
--2018-03-05 11:42:25-- https://raw.githubusercontent.com/ROBOTIS-GIT/OpenCR/develop/arduino/opencr_release/shell_update/opencr_update.tar.bz2  
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 151.101.228.133  
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|151.101.228.133|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 293362 (286K) [application/octet-stream]  
Saving to: 'opencr_update.tar.bz2'  
  
opencr_update.tar.bz2 100%[=====>] 286.49K 468KB/s in 0.6s  
  
2018-03-05 11:42:27 (468 KB/s) - 'opencr_update.tar.bz2' saved [293362/293362]  
  
opencr_update/burger_turtlebot3_core.ino.bin  
opencr_update/waffle_turtlebot3_core.ino.bin  
opencr_update/opencr_ld_shell_arm  
opencr_update/update.sh  
opencr_update/  
opencr_update/opencr_ld_shell_x86  
opencr_update/waffle.opencr  
opencr_update/burger.opencr  
opencr_update/released_1.0.17.txt  
armv7l  
arm  
OpenCR Update Start..  
opencr_ld_shell ver 1.0.0  
opencr_ld_main  
[ ] file name : burger.opencr  
[ ] file size : 172 KB  
[ ] fw_name : burger  
[ ] fw_ver : 1.0.17  
[OK] Open port : /dev/ttyACM0  
[ ]  
[ ] Board Name : OpenCR R1.0  
[ ] Board Ver : 0x17020800  
[ ] Board Rev : 0x00000000  
[OK] flash_erase : 0.92s  
[OK] flash_write : 1.84s  
[OK] CRC Check : 11A1E12 11A1E12 , 0.005000 sec  
[OK] Download  
[OK] jump_to_fw  
turtlebot@turtlebot:~$
```

Installing additional dependencies

- Copy and paste this command to your terminal:
 - `$ sudo apt-get install ros-noetic-joy ros-noetic-teleop-twist-joy \`
 - `ros-noetic-teleop-twist-keyboard ros-noetic-laser-proc \`
 - `ros-noetic-rgbd-launch ros-noetic-rosserial-arduino \`
 - `ros-noetic-rosserial-python ros-noetic-rosserial-client \`
 - `ros-noetic-rosserial-msgs ros-noetic-amcl ros-noetic-map-server \`
 - `ros-noetic-move-base ros-noetic-urdf ros-noetic-xacro \`
 - `ros-noetic-compressed-image-transport ros-noetic-rqt* ros-noetic-rviz \`
 - `ros-noetic-gmapping ros-noetic-navigation ros-noetic-interactive-markers`
- `$ cd >>> alliance` you added to your bashrc file
- `git clone https://github.com/ROBOTIS-GIT/turtlebot3_msgs.git`
- `git clone https://github.com/ROBOTIS-GIT/turtlebot3.git`
- `$ cd >>> alliance` you added to your bashrc file

Setup for .bashrc file (for remote PC)

- ifconfig >>> get your IP address
- gedit ~/.bashrc
 - At the end of the file write following commands:
 - alias eb='nano ~/.bashrc'
 - alias sb='source ~/.bashrc'
 - alias gs='git status'
 - alias gp='git pull'
 - alias cw='cd ~/catkin_ws'
 - alias cs='cd ~/catkin_ws/src'
 - alias cm='cd ~/catkin_ws && catkin_make'
 - source /opt/ros/kinetic/setup.bash
 - source ~/catkin_ws/devel/setup.bash
 - export ROS_MASTER_URI=http://**YOUR_PC_IP_ADDRESS**:11311
 - export ROS_HOSTNAME= **YOUR_PC_IP_ADDRESS**
 - Save and Exit
 - \$ source ~/.bashrc or just sb

Connecting PC with Robot

- `$ ssh pi@IP_OF_YOUR_ROBOT`
- * It depends on your robot name and Password
- `PASSWORD: pi`
- `pi@raspberrypi: “NOW YOU CONNECTED TO YOUR ROBOT WITH PC”`

Bringup

- Run command on your PC: `roscore`
- Run command on your robot: `roslaunch turtlebot3_bringup turtlebot3_robot.launch`
- Open new terminal on PC and run commands:
 - `export TURTLEBOT3_MODEL=burger >>>` every time you open new terminal
 - `roslaunch turtlebot3_bringup turtlebot3_remote.launch`

Open new terminal on PC and run

- `export TURTLEBOT3_MODEL=burger`
- `roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch`

program that can help you move your turtlebot3

Camera test

- Run ssh
 - `$ ssh pi@ip of burger`
- In the burger window of Laptop
 - `$ roslaunch turtlebot3_bringup turtlebot3_robot.launch`
- Open new terminal of Laptop
 - `$ ssh pi@ip of burger`
 - `$ roslaunch raspicam_node camerav1_1280x720.launch`
- On the burger
 - `$ roslaunch raspicam_node camerav1_1280x720.launch` (or camerav2, resolution setting)
- On the Laptop
 - `$ rqt_image_view`

Thank you

